Detecting Credit Card Fraud With a Voting Ensemble: Imbalanced Learning and Threat Intelligence

Ruthvik Nath Bandari

Northeastern University

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Mimoza Dimodugno, PhD

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Card fraud is a needle-in-a-haystack problem. In the Credit Card Fraud Detection dataset (Dal Pozzolo et al., 2015; Machine Learning Group - ULB, 2018), only 0.17% of transactions are fraudulent. This lab compared three supervised models and a voting ensemble and turned their output into threat intelligence for a bank's fraud team. At the brief's request, it also includes a short analysis of a phishing-email dataset (Cyber Cop, 2023).

Method

After 1,081 duplicate rows were removed, 283,726 transactions remained, 473 of them fraudulent. A stratified 80/20 split kept the real fraud rate in the test set. *Amount* was scaled with a robust scaler; the other 28 features are anonymized principal components. *Time* was excluded because it counts seconds from the start of the recording and would mean nothing for a live transaction. To counter the imbalance, SMOTE (Chawla et al., 2002) generated synthetic frauds up to a 1:10 ratio, only inside each cross-validation training fold, so no synthetic data reached validation or test scores.

Logistic regression, random forest, and XGBoost (Chen & Guestrin, 2016) were tuned with grid search over stratified three-fold cross-validation, selecting on F1 and reporting AUC-ROC. The best settings were C = 0.0001 for logistic regression, a maximum depth of 6 and learning rate of 0.1 for XGBoost, and fully grown trees for the random forest. The two strongest models were combined in a soft-voting ensemble. Detection thresholds were chosen from out-of-fold training predictions, never from the test set.

Results

In cross-validation, XGBoost (F1 = 0.849) and random forest (0.830) outperformed logistic regression (0.771). The XGBoost + random forest ensemble scored highest (0.854), although the gain was within one fold-to-fold standard deviation. On the test set (Table 1), the ensemble found 76 of 95 frauds with 8 false alarms (precision 0.905, F1 0.849), against 74 and 11 for random forest; with only 95 test frauds, that difference is small. AUC-ROC barely

separated the models (0.968–0.982), whereas PR-AUC did (0.672–0.829; Figure 1). PR-AUC is therefore the more informative summary.

Threat Intelligence

In the full dataset, the median fraudulent transaction was small (€9.82 against €22.00; see also the dashboard in Figure 2), although a tail of larger frauds pushed the fraud mean above the legitimate mean. The fraud rate peaked at 1.5% in hour 2 counted from the first transaction (both days pooled), one of the lowest-volume hours. Ranked by mean position across four importance methods (Figure 3), *V14*, *V12*, and *V4* were the strongest fraud indicators.

Table 2 gives the recommended thresholds. The balanced threshold (0.774) caught 70 of 95 test frauds with a precision of 0.99. The high-recall threshold caught 83 of 95, but alerts rose from 125 to 1,745 per 100,000 transactions. The default 0.5 caught 76 of 95, so thresholds need recalibration after deployment. The 25 missed frauds had a median value of €2.22 and looked legitimate on the top components, suggesting that per-card behavioral features, not a more complex model, are the next step. In a simulated real-time stream, scoring took 3.3 ms per transaction (53 ms with the uncertainty band), and uncertain transactions went to manual review.

The phishing classifier reached an AUC-ROC of 0.998, an easier task because phishing vocabulary is distinctive. With two days of 2013 data and a random split, fraud detection in production would likely be weaker.

Reflection

My Module 1 and Module 3 detectors learned only from normal traffic. This lab gave me labels, and they did not make the problem easy. With roughly one fraud in every 600 transactions, accuracy is meaningless, and SMOTE had to stay inside the cross-validation folds; applying it before splitting would have produced better-looking numbers and a worse model.

The frauds the ensemble missed looked ordinary on every component, which echoes my Module 3 autoencoder: a stronger model cannot find a signal the data does not contain. I want to build detection systems whose thresholds are set together with the analysts who handle the alerts,

and which, like the review band in my real-time scorer, know when they are unsure and hand those cases to a person.

References

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Table 1

Test-Set Performance of the Three Models and the Voting Ensemble

Model	Precision	Recall	F1	AUC-ROC	PR-AUC	False alarms
Logistic regression	0.718	0.779	0.747	0.968	0.672	29
Random forest	0.871	0.779	0.822	0.972	0.824	11
XGBoost	0.826	0.800	0.813	0.972	0.820	16
Ensemble (XGBoost + random forest)	0.905	0.800	0.849	0.982	0.829	8

Note. Test set of 56,746 transactions, 95 of them fraud, at a probability threshold of 0.5. PR-AUC = area under the precision-recall curve.

Table 2

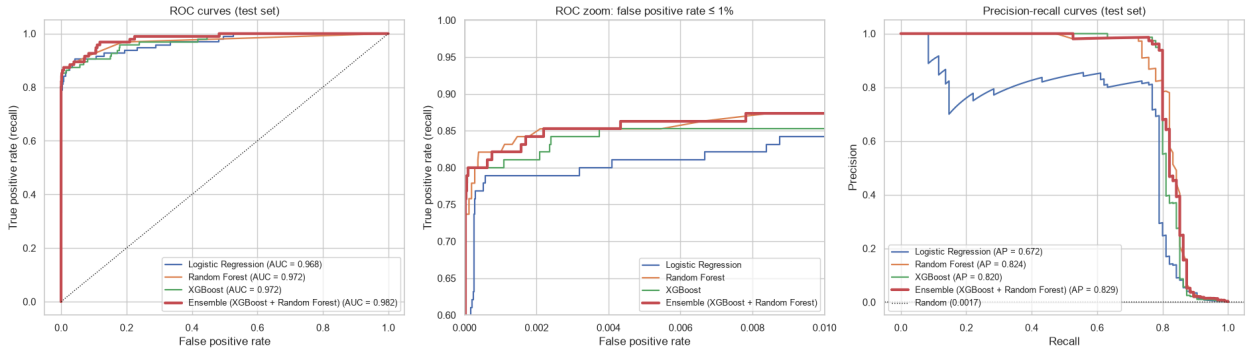
Recommended Detection Thresholds for the Ensemble

Operating point	Threshold	Frauds caught	Precision	Alerts per 100,000
Balanced (maximum F1 in training), recommended	0.774	70 of 95	0.99	125
High recall (at least 90% recall in training)	0.026	83 of 95	0.08	1,745
Alert budget (at most 100 alerts per 100,000 in training)	0.975	57 of 95	0.98	102
Reference: default threshold	0.500	76 of 95	0.90	148

Note. Thresholds were chosen from out-of-fold predictions on the training set and evaluated on the 56,746-transaction test set (95 frauds). The default threshold is shown for comparison only.

Figure 1

ROC and Precision-Recall Curves on the Test Set



Note. The middle panel zooms into false-positive rates of 1% or less. AP = average precision (PR-AUC).

Figure 2

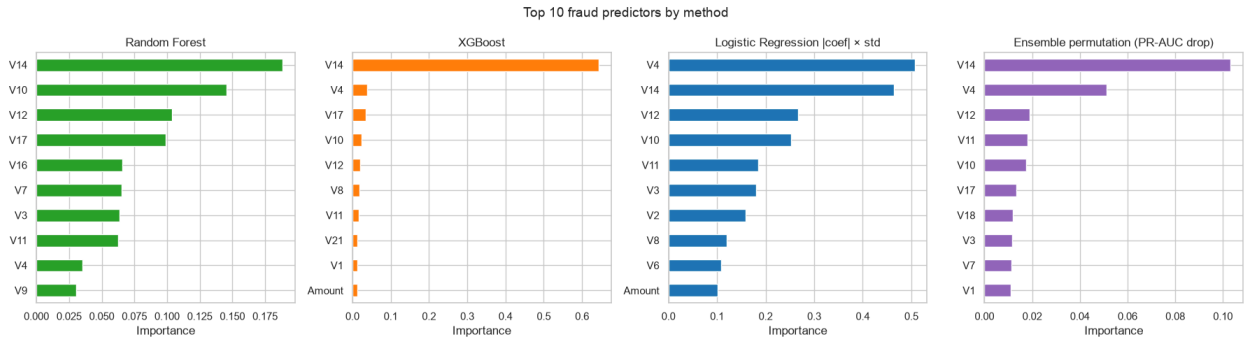
Fraud Detection Dashboard



Note. Test-set transactions scored by the ensemble at the balanced threshold of 0.774. Hours are counted from the dataset's first transaction because clock time is not provided. The 48-hour timeline uses the full dataset.

Figure 3

Top Fraud Predictors by Four Importance Methods



Note. Logistic regression coefficients are scaled by each feature's standard deviation. Permutation importance measures the drop in the ensemble's PR-AUC on the test set when a feature is shuffled.